Poisson Formula Derivation

Advanced Statistics

Letting the Number of Attempts go to Infinity

Instead of some number of attempts, n, and rate of success, p, we'll think about an expected number of successes λ and let n, the number of attempts, go to infinity (and p go to zero).

$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$

$p = \frac{\lambda}{n}$

$P(X=k) = \lim_{n\to\infty} \binom{n}{k} (\frac{\lambda}{n})^k (1-\frac{\lambda}{n})^{n-k}$

What does this distribution look like?

$P(X=k) = \lim_{n\to\infty} \binom{n}{k} (\frac{\lambda}{n})^k (1-\frac{\lambda}{n})^{n-k}$

We'll rewrite this as:

$P(X=k) = \frac{\lambda^k}{k!} \cdot [k! \cdot \binom{n}{k} (\frac{1}{n})^k] \cdot [(1-\frac{\lambda}{n})^n] \cdot [(1-\frac{\lambda}{n})^{-k}]$

and then break this into four parts:

$P(X=k) = \frac{\lambda^k}{k!} \cdot A \cdot B \cdot C$

where $A = \lim_{n\to\infty} k! \cdot \binom{n}{k} (\frac{1}{n})^k$ and $B = \lim_{n\to\infty} (1-\frac{\lambda}{n})^n$ and $C = \lim_{n\to\infty} (1-\frac{\lambda}{n})^{-k}$

Starting with A:

$A = \lim_{n\to\infty} k! \cdot \binom{n}{k} (\frac{1}{n})^k$

$\lim_{n\to\infty} k! \cdot \frac{n!}{k!(n-k)!} (\frac{1}{n})^k$

$\lim_{n\to\infty} \frac{n!}{(n-k)!} (\frac{1}{n})^k$

$\lim_{n\to\infty} n \cdot (n-1) \cdot (n-2) \cdot ... \cdot (n-k+1) (\frac{1}{n})^k$

$\lim_{n\to\infty} \frac{n \cdot (n-1) \cdot (n-2) \cdot ... \cdot (n-k+1)}{n^k}$

$\lim_{n\to\infty} \frac{n}{n} \cdot \frac{n-1}{n} \cdot \frac{n-2}{n} \cdot ... \cdot \frac{n-k+1}{n} = 1 \cdot 1 \cdot 1 \cdot ... \cdot 1 = 1$

So A just equals 1.

Now part B:

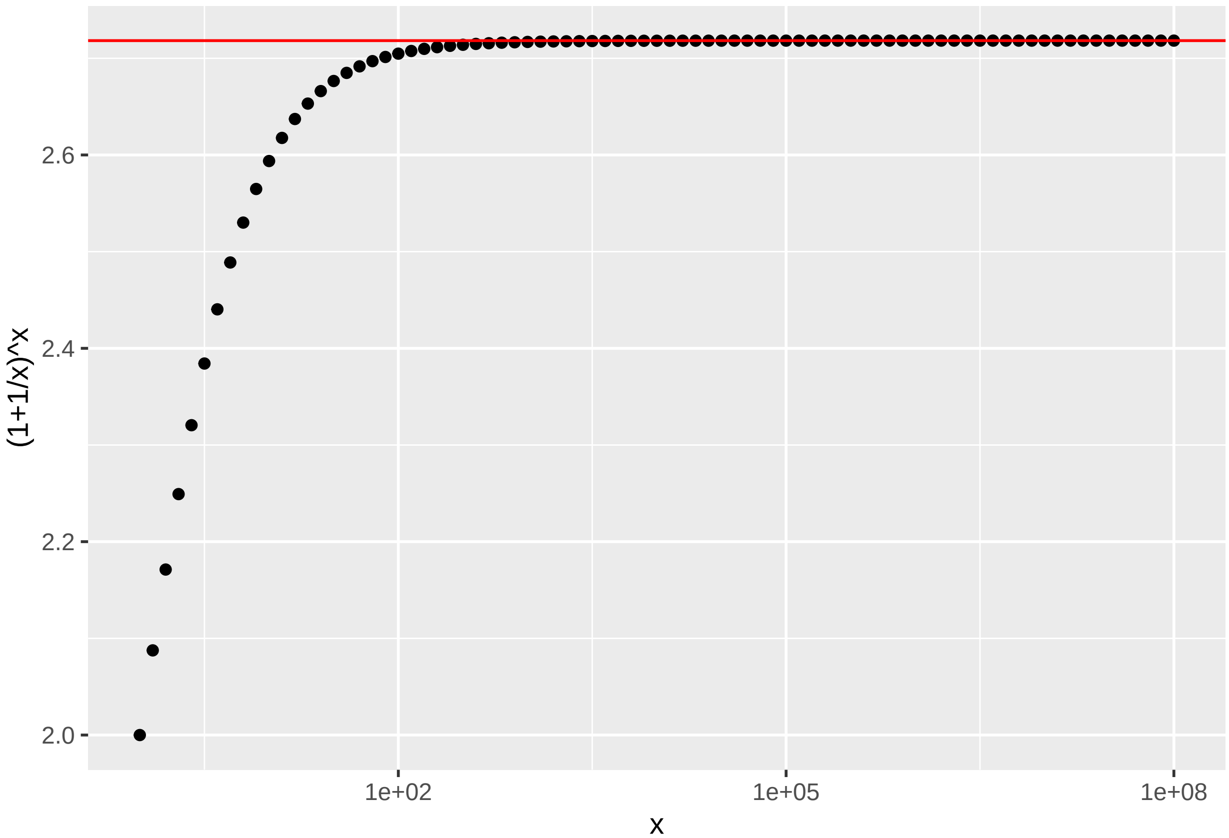
$B = \lim_{n\to\infty} (1-\frac{\lambda}{n})^n$

subtitute $j = -1 \cdot \frac{n}{\lambda}$ and we have

$B = \lim_{j\to\infty} (1+\frac{1}{j})^{-1 \cdot \lambda j} = \lim_{j\to\infty} [(1+\frac{1}{j})^j]^{-1 \cdot \lambda}$

But what is $\lim_{j \to \infty}(1 + \frac{1}{j})^j$?

Euler's Number



Finally, part C:

So $B = e^{-\lambda}$

What about C? Recall, $C = \lim_{n \to \infty}(1 - \frac{\lambda}{n})^{-k}$.

and as $n \to \infty$ this becomes 1^{-k} or $\frac{1}{1^k}$ or just 1.

Putting it all together we have:

$P(X = k) = \frac{\lambda^k}{k!} \cdot 1 \cdot e^{-\lambda} \cdot 1$

$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$